

DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY

MODEL QUESTION PAPER

ORGANIC TECHNOLOGY

(MAXIMUM MARKS 100)

Time -3 hours

PART-A

(Answer all questions in one or two sentences. Each questions carries 2 marks)

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| I | | 2 |
| 1 | Define Iodine value of oil | |
| 2 | Define rancidity | 2 |
| 3 | What is black liquor | 2 |
| 4 | Define sensitivity of an explosive | 2 |
| 5 | What is chrome tanning | 2 |

PART-B

(Answer any five questions .Each questions carries 6 marks)

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|----|--|---|
| II | | 6 |
| 1 | Explain hydrogenation process of oil. | |
| 2 | Explain the batch process of soap manufacture. | 6 |
| 3 | Describe the manufacture of laminated paper. | 6 |
| 4 | Describe the preparation of pulp for paper making. | 6 |
| 5 | Explain the manufacture of smokeless powder. | 6 |
| 6 | Explain the refining process of sugar. | 6 |
| 7 | Describe the finishing operations of leather. | 6 |

PART-C

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| III | <u>MODULE-1</u> | |
| a | Draw a neat sketch and explain oil extraction by Ballmann extractor. | 10 |
| b | Describe the mechanism of detergents. | 5 |
| | OR | |
| IV | a Explain the manufacture of laundry soap from neat soap with flow diagram. | 10 |
| | b Describe the bleaching process used in oil manufacture. | 5 |

MODULE-II

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| V | a Explain the manufacture of paper by four driner paper machine with a neat sketch. | 12 |
| | b Describe the manufacture of pulp from waste water. | 3 |
| | OR | |
| VI | a Describe sulphate process used for pulp manufacture with a flow diagram. | 12 |

- b Write soda process for pulping. 3

MODULE-III

- VII a Differentiate between military and commercial explosive 8
b Describe the process of high explosive ammunition. 7
OR
VIII a Explain the manufacture of bio insecticide neem cow urine extract and mention its 7
applications.
b Describe the manufacture of lead oxide. 8

MODULE-IV

- IX a Explain the manufacture of starch from tapioca. 7
b Describe vegetable tanning and chrome tanning process for making leather. 8
OR
X a Describe preparation of skin and hide for tanning. 8
b Describe evaporation and crystallization cane juice for manufacturing sugar. 7